

DEPARTMENT OF MECHANICAL ENGINEERING

Dynamics of Machinery

Unit - I D'Alembert's Principle-Inertia forces of reciprocating parts, Dynamic analysis of four link and slider-crank mechanisms, Engine force Analysis Turning moment on crankshaft, Dynamically Equivalent system, Inertia forces in a reciprocating engine, Turning Moment diagrams, Fluctuations of Energy and speed, Flywheel. (12 hours)

Unit - II Basic concepts of S.H.M, Causes and effects of vibration and degrees of freedom. Natural frequency of free oscillations – equivalent system – energy method – simple problems, damped free vibration of single degree of freedom system, forced vibration. Basic of vibration isolation, Transmissibility and vibration absorbers. (12 hours)

Unit - III Transverse vibrations of beams-Natural frequency by energy method, Dunkerly's method, Whirling of shafts- calculation of whirling speed for loaded shafts. Torsional vibrations-causes of Torsional vibration. Torsional Vibration of two and three rotor systems. Equivalent shaft system, Geared system. (12 hours)

Unit - IV Governors - Types - Centrifugal governors - Gravity controlled and spring controlled centrifugal governors –Characteristics - Effect of friction - Controlling Force - other Governor mechanisms. Gyroscopes - Gyroscopic forces and Torques -Gyroscopic stabilization - Gyroscopic effects in Automobiles, ships and airplanes (12 hours)

Unit - V Static and dynamic balancing of rotating masses in different planes - partial balancing of reciprocating masses of inline, V, W and radial engines. (12 hours)

Text Books:

1. J.E.Shigley and J.J.Uicker - Theory of Machines & Mechanisms, McGraw Hill International Edition, 2006.
2. Rattan - Theory of Machines, Tata McGraw Hill, 2009.

Reference Books:

3. J.S.Rao and R.V.Dukkipati - Mechanism and Machine Theory, New Age International, 2004.
4. Thomas Bevan - Theory of Machines, CBS Publishers & Distributors, 2004.
5. P.L.Ballaney - Mechanics of Machines, Khanna Publishers, 2005.
6. Robert F.Steidel Jr. - An introduction to Mechanical Vibrations, John Wiley & Sons Inc.,New York, 2003.

Lesson Modules

	UNIT – I			
Modules		Lectures	Content	Hours
Module 1	Inertia Forces	Lecture 1	D’Alembert’s Principle	1
		Lecture 2	Inertia forces and Turning moment on a parts of a reciprocating engine	1
		Lecture 3	Problems	1
Module 2	Dynamic Analysis	Lecture 4	Dynamic analysis of four link and slider-crank mechanisms and	1
		Lecture 5	Problems	1
Module 3	Equivalent system	Lecture 6	Dynamically Equivalent system	1
		Lecture 7	Problems	2
Module 4	Turning Moment	Lecture 8	Turning Moment diagrams	1
		Lecture 9	Fluctuations of Energy and speed and Flywheel	1
		Lecture 10	Problems	2
			Total No of Hours	12

Module 1	Inertia Forces
Lecture 1	D'Alembert's Principle

D'Alembert's Principle

Inertia Forces

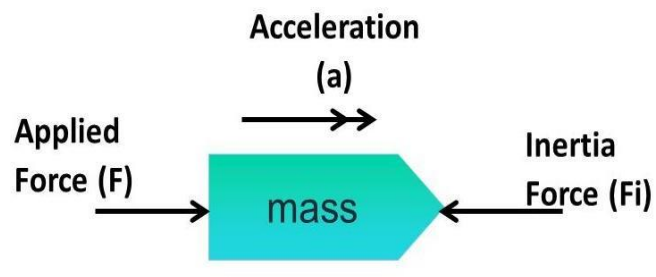
- The inertia force is an *imaginary* force.
- Which acts upon a rigid body, brings it in an equilibrium position.
- It is numerically equal to the accelerating force in *magnitude* and *Opposite in direction*.
- Mathematically,
- Inertia force = – Accelerating force = – $m.a$
- Where,

m = Mass of the body, and

a = Linear acceleration of the centre of gravity of the body.

- Similarly, the inertia torque is an imaginary torque.
- Which when applied upon the rigid body, brings it in equilibrium position.
- It is equal to the accelerating couple in magnitude but *opposite in direction*.

Consider a rigid body acted upon by a system of forces. The system may be reduced to a single resultant force acting on the body whose magnitude is given by the product of the mass of the body and the linear acceleration of the centre of mass of the body. According to Newton's second law of motion,



$$F = m.a \quad \dots(i)$$

where

F = Resultant force acting on the body,

m = Mass of the body, and

a = Linear acceleration of the centre of mass of the body.

The equation (i) may also be written as:

$$F - m.a = 0 \quad \dots(ii)$$

A little consideration will show, that if the quantity $-m.a$ be treated as a force, equal, opposite and with the same line of action as the resultant force F , and include this force with the system of forces of which F is the resultant, then the complete system of forces will be in equilibrium. This principle is known as **D'Alembert's principle**. The equal and opposite force $-m.a$ is known as **reversed effective force** or the **inertia force** (briefly written as F_i). The equation (ii) may be written as

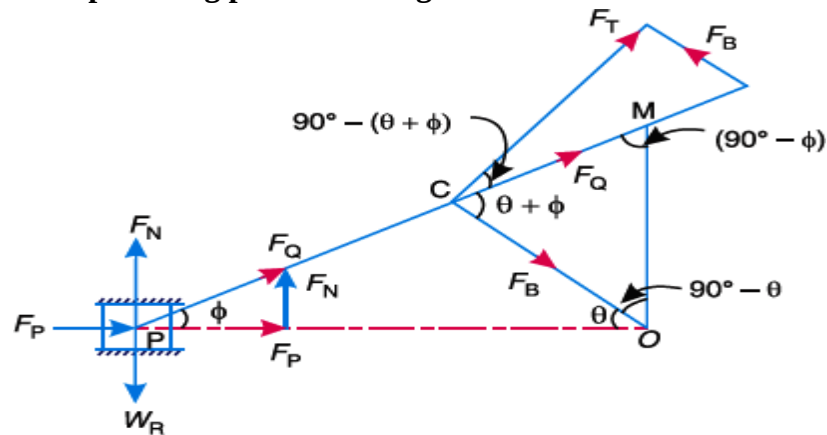
$$F + F_i = 0 \quad \dots(iii)$$

Thus, D'Alembert's principle states that **the resultant force acting on a body together with the reversed effective force (or inertia force), are in equilibrium.**

This principle is used to reduce a dynamic problem into an equivalent static problem.

Module 1	Inertia Forces
Lecture 2	Inertia forces and Turning moment on a parts of a reciprocating engine

Inertia forces on the reciprocating parts of an engine



Forces on the reciprocating parts of an engine.

1. Piston effort.

- It is the net force acting on the piston or crosshead pin, along the line of stroke. It is denoted by F_P in Fig.

- The acceleration of the reciprocating parts,

$$a_R = a_P = \omega^2 \cdot r \left(\cos \theta + \frac{\cos 2\theta}{n} \right)$$

- Accelerating force or inertia force of the reciprocating parts,

$$F_I = m_R \cdot a_R = m_R \cdot \omega^2 \cdot r \left(\cos \theta + \frac{\cos 2\theta}{n} \right)$$

- Therefore,

- Piston effort, $F_P = \text{Net load on the piston} \pm \text{Inertia force}$
 $= F_L \pm F_I \dots (\text{Neglecting frictional resistance})$
 $= F_L \pm F_I - R_F \dots (\text{Considering frictional resistance})$

where, $R_F = \text{Frictional resistance}$

- The -ve sign is used when the piston is accelerated, and +ve sign is used when the piston is retarded.
- In a reciprocating steam engine, net load on the piston,

$$F_L = p_1 A_1 - p_2 A_2 = p_1 A_1 - p_2 (A_1 - a)$$

- Where,

$p_1, A_1 = \text{Pressure and cross-sectional area on the back end side of the piston,}$

$p_2, A_2 = \text{Pressure and cross-sectional area on the crank end side of the piston,}$

$a = \text{Cross-sectional area of the piston rod.}$

Notes :

- If 'p' is the net pressure of steam or gas on the piston and D is diameter of the piston, then

$$\text{Net load on the piston, } F_L = \text{Pressure} \times \text{Area} = p \times \frac{\pi}{4} \times D^2$$

- In case of a vertical engine,

$$\text{Piston effort, } F_P = F_L \mp F_I \pm W_R - R_F$$

2. Force acting along the connecting rod.

- It is denoted by F_Q in Fig. From the geometry of the figure, we find that

$$F_Q = \frac{F_P}{\cos \phi}$$

$$\text{We know that } \cos \phi = \sqrt{1 - \frac{\sin^2 \theta}{n^2}}$$

$$\therefore F_Q = \frac{F_P}{\sqrt{1 - \frac{\sin^2 \theta}{n^2}}}$$

3. Thrust on the sides of the cylinder walls or normal reaction on the guide bars

- It is denoted by F_N in Fig. From the figure, we find that

$$F_N = F_Q \sin \phi = \frac{F_P}{\cos \phi} \times \sin \phi = F_P \tan \phi \quad \dots \left[\because F_Q = \frac{F_P}{\cos \phi} \right]$$

4. Crank-pin effort and thrust on crank shaft bearings.

- The force acting on the connecting rod F_Q may be resolved into two components, one perpendicular to the crank and the other along the crank.
- The component of F_Q perpendicular to the crank is known as crank-pin effort and it is denoted by F_T in Fig.
- The component of F_Q along the crank produces a thrust on the crank shaft bearings and it is denoted by F_B in Fig.
- Resolving F_Q perpendicular to the crank,

$$F_T = F_Q \sin (\theta + \phi) = \frac{F_P}{\cos \phi} \times \sin (\theta + \phi)$$

- and resolving F_Q along the crank,

$$F_B = F_Q \cos (\theta + \phi) = \frac{F_P}{\cos \phi} \times \cos (\theta + \phi)$$

5. Crank effort or turning moment or torque on the crank shaft.

- The product of the crankpin effort (F_T) and the crank pin radius (r) is known as crank effort or turning moment or torque on the crank shaft.
- Mathematically, Crank effort,

$$\begin{aligned} T &= F_T \times r = \frac{F_P \sin (\theta + \phi)}{\cos \phi} \times r \\ &= \frac{F_P (\sin \theta \cos \phi + \cos \theta \sin \phi)}{\cos \phi} \times r \\ &= F_P \left(\sin \theta + \cos \theta \times \frac{\sin \phi}{\cos \phi} \right) \times r \\ &= F_P (\sin \theta + \cos \theta \tan \phi) \times r \quad \dots (i) \end{aligned}$$

We know that $l \sin \phi = r \sin \theta$

$$\sin \phi = \frac{r}{l} \sin \theta = \frac{\sin \theta}{n} \quad \dots \left(\because n = \frac{l}{r} \right)$$

and $\cos \phi = \sqrt{1 - \sin^2 \phi} = \sqrt{1 - \frac{\sin^2 \theta}{n^2}} = \frac{1}{n} \sqrt{n^2 - \sin^2 \theta}$

$$\therefore \tan \phi = \frac{\sin \phi}{\cos \phi} = \frac{\sin \theta}{n} \times \frac{n}{\sqrt{n^2 - \sin^2 \theta}} = \frac{\sin \theta}{\sqrt{n^2 - \sin^2 \theta}}$$

Substituting the value of $\tan \phi$ in equation (i), we have crank effort,

$$\begin{aligned} T &= F_P \left(\sin \theta + \frac{\cos \theta \sin \theta}{\sqrt{n^2 - \sin^2 \theta}} \right) \times r \\ &= F_P \times r \left(\sin \theta + \frac{\sin 2\theta}{2\sqrt{n^2 - \sin^2 \theta}} \right) \quad \dots (ii) \end{aligned}$$

$\dots (\because 2 \cos \theta \sin \theta = \sin 2\theta)$

Note: Since $\sin^2 \theta$ is very small as compared to n^2 therefore neglecting $\sin^2 \theta$, we have,

Crank effort, $T = F_P \times r \left(\sin \theta + \frac{\sin 2\theta}{2n} \right) = F_P \times OM$

We have seen in Art. 15.8, that

$$OM = r \left(\sin \theta + \frac{\sin 2\theta}{2n} \right)$$

Therefore, it is convenient to find OM instead of solving the large expression.

Module 1	Inertia Forces
Lecture 3	Problems on Inertia forces

Problems:

- Find the inertia force for the following data of an I.C. engine. Bore = 175 mm, stroke = 200 mm, engine speed = 500 r.p.m., length of connecting rod = 400 mm, crank angle = 60° from T.D.C and mass of reciprocating parts = 180 kg.

We know that ratio of lengths of connecting rod and crank,

$$n = l / r = 0.4 / 0.1 = 4$$

$$\begin{aligned} \therefore \text{Inertia force, } F_I &= m_R \cdot \omega^2 r \left(\cos \theta + \frac{\cos 2\theta}{n} \right) \\ &= 180 \times (52.4)^2 \times 0.1 \left(\cos 60^\circ + \frac{\cos 120^\circ}{4} \right) = 18\,530 \text{ N} \end{aligned}$$

$$= 18.53 \text{ kN Ans.}$$

- The crank-pin circle radius of a horizontal engine is 300 mm. The mass of the reciprocating parts is 250 kg. When the crank has travelled 60° from I.D.C., the difference between the driving and the back pressures is 0.35 N/mm². The connecting rod length between centers is 1.2 m and the cylinder bore is 0.5 m. If the engine runs at 250 r.p.m. and if the effect of piston rod diameter is neglected, calculate: **1.** pressure on slide bars, **2.** thrust in the connecting rod, **3.** tangential force on the crank-pin, and **4.** turning moment on the crank shaft.

Solution. Given: $r = 300 \text{ mm} = 0.3 \text{ m}$; $m_R = 250 \text{ kg}$; $\theta = 60^\circ$; $p_1 - p_2 = 0.35 \text{ N/mm}^2$; $l = 1.2 \text{ m}$; $D = 0.5 \text{ m} = 500 \text{ mm}$; $N = 250 \text{ r.p.m.}$ or $\omega = 2\pi \times 250/60 = 26.2 \text{ rad/s}$

First of all, let us find out the piston effort (F_P).

We know that net load on the piston,

$$F_L = (p_1 - p_2) \frac{\pi}{4} \times D^2 = 0.35 \times \frac{\pi}{4} (500)^2 = 68\,730 \text{ N}$$

...($\because$ Force = Pressure $\times$ Area)

Ratio of length of connecting rod and crank,

$$n = l / r = 1.2 / 0.3 = 4$$

and accelerating or inertia force on reciprocating parts,

$$\begin{aligned} F_I &= m_R \cdot \omega^2 r \left(\cos \theta + \frac{\cos 2\theta}{n} \right) \\ &= 250 (26.2)^2 \cdot 0.3 \left(\cos 60^\circ + \frac{\cos 120^\circ}{4} \right) = 19\,306 \text{ N} \end{aligned}$$

$$\therefore \text{Piston effort, } F_P = F_L - F_I = 68\,730 - 19\,306 = 49\,424 \text{ N} = 49.424 \text{ kN}$$

1. Pressure on slide bars

Let ϕ = Angle of inclination of the connecting rod to the line of stroke.

$$\text{We know that, } \sin \phi = \frac{\sin \theta}{n} = \frac{\sin 60^\circ}{4} = \frac{0.866}{4} = 0.2165$$

$$\therefore \phi = 12.5^\circ$$

We know that pressure on the slide bars,

$$F_N = F_P \tan \phi = 49.424 \times \tan 12.5^\circ = 10.96 \text{ kN Ans.}$$

2. Thrust in the connecting rod

We know that thrust in the connecting rod,

$$F_Q = \frac{F_P}{\cos \phi} = \frac{49.424}{\cos 12.5^\circ} = 50.62 \text{ kN Ans.}$$

3. Tangential force on the crank-pin

We know that tangential force on the crank pin,

$$F_T = F_Q \sin (\theta + \phi) = 50.62 \sin (60^\circ + 12.5^\circ) = 48.28 \text{ kN Ans.}$$

4. Turning moment on the crank shaft

We know that turning moment on the crank shaft,

$$T = F_T \times r = 48.28 \times 0.3 = 14.484 \text{ kN-m Ans.}$$

3. A vertical double acting steam engine has a cylinder 300 mm diameters and 450 mm stroke and runs at 200 r.p.m. The reciprocating part has a mass of 225 kg and the piston rod is 50 mm diameter. The connecting rod is 1.2 m long. When the crank has turned through 125° from the top dead centre, the steam pressure above the piston is 30 kN/m^2 and below the piston is 1.5 kN/m^2 . Calculate the effective turning moment on the crank shaft.

Solution. Given : $D = 300 \text{ mm} = 0.3 \text{ m}$; $L = 450 \text{ mm}$ or $r = L/2 = 225 \text{ mm} = 0.225 \text{ m}$;
 $N = 200 \text{ r.p.m.}$ or $\omega = 2\pi \times 200/60 = 20.95 \text{ rad/s}$; $m_R = 225 \text{ kg}$; $d = 50 \text{ mm} = 0.05 \text{ m}$; $l = 1.2 \text{ m}$;
 $\theta = 125^\circ$; $p_1 = 30 \text{ kN/m}^2 = 30 \times 10^3 \text{ N/m}^2$; $p_2 = 1.5 \text{ kN/m}^2 = 1.5 \times 10^3 \text{ N/m}^2$

We know that area of the piston,

$$A_1 = \frac{\pi}{4} \times D^2 = \frac{\pi}{4} \times (0.3)^2 = 0.0707 \text{ m}^2$$

and area of the piston rod, $a = \frac{\pi}{4} \times d^2 = \frac{\pi}{4} \times (0.05)^2 = 0.00196 \text{ m}^2$

$\therefore$ Force on the piston due to steam pressure,

$$\begin{aligned} F_L &= p_1 A_1 - p_2 (A_1 - a) \\ &= 30 \times 10^3 \times 0.0707 - 1.5 \times 10^3 (0.0707 - 0.00196) \text{ N} \\ &= 2121 - 103 = 2018 \text{ N} \end{aligned}$$

Ratio of lengths of connecting rod and crank,

$$n = l/r = 1.2/0.225 = 5.33$$

and inertia force on the reciprocating parts,

$$\begin{aligned} F_I &= m_R \cdot \omega^2 \cdot r \left(\cos \theta + \frac{\cos 2\theta}{n} \right) \\ &= 225 (20.95)^2 \times 0.225 \left(\cos 125^\circ + \frac{\cos 250^\circ}{5.33} \right) = -14172 \text{ N} \end{aligned}$$

We know that for a vertical engine, net force on the piston or piston effort,

$$\begin{aligned} F_P &= F_L - F_I + m_R g \\ &= 2018 - (-14172) + 225 \times 9.81 = 18397 \text{ N} \end{aligned}$$

Let ϕ = Angle of inclination of the connecting rod to the line of stroke.

We know that, $\sin \phi = \frac{\sin \theta}{n} = \frac{\sin 125^\circ}{5.33} = \frac{0.8191}{5.33} = 0.1537$ $\phi = 8.84^\circ$

We know that effective turning moment on the crank shaft,

$$T = \frac{F_P \times \sin (\theta + \phi)}{\cos \phi} \times r = \frac{18397 \sin (125^\circ + 8.84^\circ)}{\cos 8.84^\circ} \times 0.225 \text{ N-m} = 3021.6 \text{ N-m Ans.}$$

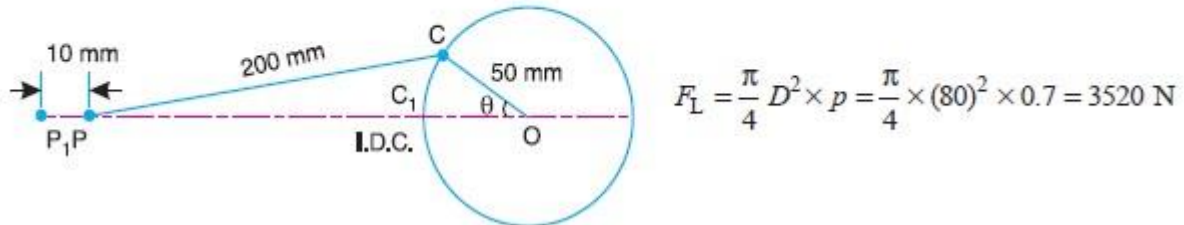
4. The crank and connecting rod of a petrol engine, running at 1800 r.p.m. are 50 mm and 200 mm respectively. The diameter of the piston is 80 mm and the mass of the reciprocating parts is 1 kg. At a point during the power stroke, the pressure on the piston is 0.7 N/mm^2 , when it has moved 10 mm from

the inner dead centre. Determine: **1.** Net load on the gudgeon pin, **2.** Thrust in the connecting rod, **3.** Reaction between the piston and cylinder, and **4.** The engine speed at which the above values become zero.

Solution. Given : $N = 1800$ r.p.m. or $\omega = 2\pi \times 1800/60 = 188.52$ rad/s ; $r = 50$ mm = 0.05 m ;
 $l = 200$ mm ; $D = 80$ mm ; $m_R = 1$ kg ; $p = 0.7$ N/mm² ; $x = 10$ mm

1. Net load on the gudgeon pin

We know that load on the piston,



When the piston has moved 10 mm from the inner dead centre, i.e. when $P_1P = 10$ mm, the crank rotates from OC_1 to OC through an angle θ as shown in Fig. 15.10.

By measurement, we find that $\theta = 33^\circ$.

We know that ratio of lengths of connecting rod and crank,

$$n = l/r = 200/50 = 4$$

and inertia force on the reciprocating parts,

$$\begin{aligned} F_I &= m_R \cdot a_R = m_R \cdot \omega^2 \cdot r \left(\cos \theta + \frac{\cos 2\theta}{n} \right) \\ &= 1 \times (188.52)^2 \times 0.05 \left(\cos 33^\circ + \frac{\cos 66^\circ}{4} \right) = 1671 \text{ N} \end{aligned}$$

We know that net load on the gudgeon pin,

$$F_P = F_L - F_I = 3520 - 1671 = 1849 \text{ N Ans.}$$

2. Thrust in the connecting rod

Let

ϕ = Angle of inclination of the connecting rod to the line of stroke.

We know that,

$$\sin \phi = \frac{\sin \theta}{n} = \frac{\sin 33^\circ}{4} = \frac{0.5446}{4} = 0.1361$$

$$\therefore \phi = 7.82^\circ$$

We know that thrust in the connecting rod,

$$F_Q = \frac{F_P}{\cos \phi} = \frac{1849}{\cos 7.82^\circ} = 1866.3 \text{ N Ans.}$$

3. Reaction between the piston and cylinder

We know that reaction between the piston and cylinder,

$$F_N = F_P \tan \phi = 1849 \tan 7.82^\circ = 254 \text{ N Ans.}$$

4. Engine speed at which the above values will become zero

A little consideration will show that the above values will become zero, if the inertia force on the reciprocating parts (F_I) is equal to the load on the piston (F_L). Let ω_1 be the speed in rad/s, at which $F_I = F_L$.

$$\therefore m_R (\omega_1)^2 r \left(\cos \theta + \frac{\cos 2\theta}{n} \right) = \frac{\pi}{4} D^2 \times p$$

$$1 (\omega_1)^2 \times 0.05 \left(\cos 33^\circ + \frac{\cos 66^\circ}{4} \right) = \frac{\pi}{4} \times (80)^2 \times 0.7 \quad \text{or} \quad 0.047 (\omega_1)^2 = 3520$$

$$\therefore (\omega_1)^2 = 3520 / 0.047 = 74894 \quad \text{or} \quad \omega_1 = 273.6 \text{ rad/s}$$

$\therefore$ Corresponding speed in r.p.m.,

$$N_1 = 273.6 \times 60 / 2\pi = 2612 \text{ r.p.m. Ans.}$$

5. During a trial on steam engine, it is found that the acceleration of the piston is 36 m/s² when the crank has moved 30° from the inner dead centre position. The net effective steam pressure on the piston is 0.5 N/mm² and the frictional resistance is equivalent to a force of 600 N. The diameter of the piston is 300 mm and the mass of the reciprocating parts is 180 kg. If the length of the crank is 300 mm and the ratio of the connecting rod length to the crank length is 4.5, find: **1.** Reaction on the guide bars, **2.** Thrust on the crank shaft bearings, and **3.** Turning moment on the crank shaft.

Solution. Given : $a_p = 36 \text{ m/s}^2$; $\theta = 30^\circ$; $p = 0.5 \text{ N/mm}^2$; $R_F = 600 \text{ N}$; $D = 300 \text{ mm}$; $m_R = 180 \text{ kg}$; $r = 300 \text{ mm} = 0.3 \text{ m}$; $n = l/r = 4.5$

1. Reaction on the guide bars

First of all, let us find the piston effort (F_P). We know that load on the piston,

$$F_L = p \times \frac{\pi}{4} \times D^2 = 0.5 \times \frac{\pi}{4} \times (300)^2 = 35350 \text{ N}$$

and inertia force due to reciprocating parts,

$$F_I = m_R \times a_p = 180 \times 36 = 6480 \text{ N}$$

$$\therefore \text{Piston effort, } F_P = F_L - F_I - R_F = 35350 - 6480 - 600 = 28270 \text{ N} = 28.27 \text{ kN}$$

Let ϕ = Angle of inclination of the connecting rod to the line of stroke.

$$\text{We know that } \sin \phi = \sin \theta / n = \sin 30^\circ / 4.5 = 0.1111$$

$$\therefore \phi = 6.38^\circ$$

We know that reaction on the guide bars,

$$F_N = F_P \tan \phi = 28.27 \tan 6.38^\circ = 3.16 \text{ kN Ans.}$$

2. Thrust on the crank shaft bearing

We know that thrust on the crank shaft bearings,

$$F_B = \frac{F_P \cos(\theta + \phi)}{\cos \phi} = \frac{28.27 \cos(30^\circ + 6.38^\circ)}{\cos 6.38^\circ} = 22.9 \text{ kN Ans.}$$

3. Turning moment on the crank shaft

We know that turning moment on the crank shaft,

$$\begin{aligned} T &= \frac{F_P \sin(\theta + \phi)}{\cos \phi} \times r = \frac{28.27 \sin(30^\circ + 6.38^\circ)}{\cos 6.38^\circ} \times 0.3 \text{ kN-m} \\ &= 5.06 \text{ kN-m} \end{aligned}$$

Module 2	Dynamic Analysis
Lecture 4	Dynamic analysis of four link and slider-crank mechanisms

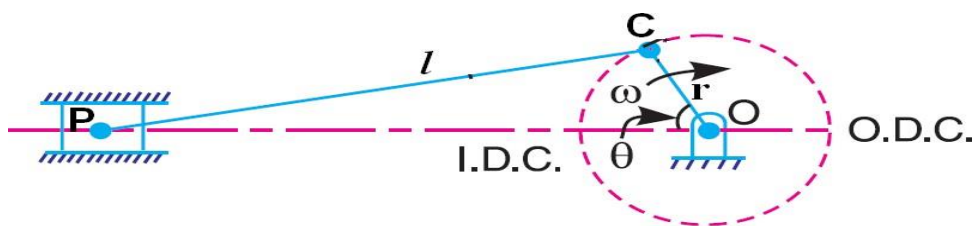
Dynamic analysis of four link and slider-crank mechanisms

- The study of forces in IC engine mechanism are classified into two types as follows :
 - Steady force analysis,
 - Dynamic force analysis.
- In static force analysis, we do not consider the effect of inertia forces arising due to the mass of the connecting rod.
- In dynamic force analysis, we also consider the effect of inertia forces caused due to the mass of connecting rod.
- The force analysis can be done both by analytical and graphical methods.

Dynamic Analysis of Four bar Mechanism:

A four-bar linkage or simply a 4-bar or four-bar is the simplest movable linkage. It consists of four rigid bodies (called bars or links), each attached to two others by single joints or pivots to form a closed loop. Four-bars are simple mechanisms common in mechanical engineering machine design and fall under the study of kinematics. Dynamic Analysis of Reciprocating engines. Inertia force and torque analysis by neglecting weight of connecting rod. Velocity and acceleration of piston. Angular velocity and Angular acceleration of connecting rod. Force and Torque Analysis in reciprocating engine neglecting the weight of connecting rod. Equivalent Dynamical System Determination of two masses of equivalent dynamical system

Dynamic Analysis of slider-crank mechanisms



Slider crank mechanism.

- Consider an IC engine mechanism shown in Fig. in which the crank OC rotates at an angular speed of ω in clockwise direction.
- At an instant it is inclined at angle θ from I.D.C.

Let,

r = Radius of crank,

l = Length of connecting rod,

n = Obliquity ratio = l/r

m_R = Mass of the reciprocating parts, e.g. piston, crosshead pin or gudgeon pin etc., in

kg,

W_R = Weight of the reciprocating parts in Newton's = $m_R \cdot g$

m = Mass of the connecting rod,

θ = Angle made by crank with I.D.C.,

Φ = Angle made by connecting rod with line of reciprocation (i.e. line OP) when line is inclined at angle θ

Module 2	Dynamic Analysis
Lecture 5	Problems on Dynamic analysis

Problems:

1. If the crank and the connecting rod are 300 mm and 1 m long respectively and the crank rotates at a constant speed of 200 r.p.m., determine: **(a)** The crank angle at which the maximum velocity occurs and **(b)** Maximum velocity of the piston.

Solution. Given : $r = 300 \text{ mm} = 0.3 \text{ m}$; $l = 1 \text{ m}$; $N = 200 \text{ r.p.m.}$ or $\omega = 2\pi \times 200/60 = 20.95 \text{ rad/s}$

1. Crank angle at which the maximum velocity occurs

Let θ = Crank angle from the inner dead centre at which the maximum velocity occurs.

We know that ratio of length of connecting rod to crank radius,

$$n = l/r = 1/0.3 = 3.33$$

and velocity of the piston,

$$v_p = \omega r \left(\sin \theta + \frac{\sin 2\theta}{2n} \right) \quad \dots(i)$$

For maximum velocity of the piston,

$$\frac{dv_p}{d\theta} = 0 \quad \text{i.e.} \quad \omega r \left(\cos \theta + \frac{2 \cos 2\theta}{2n} \right) = 0$$

$$\text{or} \quad n \cos \theta + 2 \cos^2 \theta - 1 = 0 \quad \dots(\because \cos 2\theta = 2 \cos^2 \theta - 1)$$

$$2 \cos^2 \theta + 3.33 \cos \theta - 1 = 0$$

$$\therefore \cos \theta = \frac{-3.33 \pm \sqrt{(3.33)^2 + 4 \times 2 \times 1}}{2 \times 2} = 0.26 \quad \dots(\text{Taking + ve sign})$$

$$\text{or} \quad \theta = 75^\circ \text{ Ans.}$$

2. Maximum velocity of the piston

Substituting the value of $\theta = 75^\circ$ in equation (i), maximum velocity of the piston,

$$\begin{aligned} v_{p(max)} &= \omega r \left[\sin 75^\circ + \frac{\sin 150^\circ}{2n} \right] = 20.95 \times 0.3 \left[0.966 + \frac{0.5}{3.33} \right] \text{ m/s} \\ &= 6.54 \text{ m/s Ans.} \end{aligned}$$

2. The crank and connecting rod of a steam engine are 0.3 m and 1.5 m in length. The crank rotates at 180 r.p.m. clockwise. Determine the velocity and acceleration of the piston when the crank is at 40 degrees from the inner dead centre position. Also determine the position of the crank for zero acceleration of the piston.

Solution. Given : $r = 0.3$; $l = 1.5 \text{ m}$; $N = 180 \text{ r.p.m.}$ or $\omega = \pi \times 180/60 = 18.85 \text{ rad/s}$; $\theta = 40^\circ$

Velocity of the piston

We know that ratio of lengths of the connecting rod and crank,

$$n = l/r = 1.5/0.3 = 5$$

$\therefore$ Velocity of the piston,

$$\begin{aligned} v_p &= \omega r \left(\sin \theta + \frac{\sin 2\theta}{2n} \right) = 18.85 \times 0.3 \left(\sin 40^\circ + \frac{\sin 80^\circ}{2 \times 5} \right) \text{ m/s} \\ &= 4.19 \text{ m/s Ans.} \end{aligned}$$

Acceleration of the piston

We know that acceleration of piston,

$$a_p = \omega^2 r \left(\cos \theta + \frac{\cos 2\theta}{n} \right) = (18.85)^2 \times 0.3 \left(\cos 40^\circ + \frac{\cos 80^\circ}{5} \right) \text{ m/s}^2$$
$$= 85.35 \text{ m/s}^2 \text{ Ans.}$$

Position of the crank for zero acceleration of the piston

Let θ_1 = Position of the crank from the inner dead centre for zero acceleration of the piston.

We know that acceleration of piston,

$$a_p = \omega^2 r \left(\cos \theta_1 + \frac{\cos 2\theta_1}{n} \right) \quad 0 = \frac{\omega^2 r}{n} (n \cos \theta_1 + \cos 2\theta_1)$$

$$\therefore n \cos \theta_1 + \cos 2\theta_1 = 0 \quad 5 \cos \theta_1 + 2 \cos^2 \theta_1 - 1 = 0$$

$$2 \cos^2 \theta_1 + 5 \cos \theta_1 - 1 = 0 \quad \theta_1 = 79.27^\circ \text{ or } 280.73^\circ \text{ Ans.}$$

3. In a slider crank mechanism, the length of the crank and connecting rod are 150 mm and 600 mm respectively. The crank position is 60° from inner dead centre. The crank shaft speed is 450 r.p.m. (clockwise). Using analytical method, determine: 1. Velocity and acceleration of the slider, and 2. Angular velocity and angular acceleration of the connecting rod.

Solution. Given : $r = 150 \text{ mm} = 0.15 \text{ m}$; $l = 600 \text{ mm} = 0.6 \text{ m}$; $\theta = 60^\circ$; $N = 400 \text{ r.p.m}$ or $\omega = \pi \times 450/60 = 47.13 \text{ rad/s}$

1. Velocity and acceleration of the slider

We know that ratio of the length of connecting rod and crank,

$$n = l / r = 0.6 / 0.15 = 4$$

$\therefore$ Velocity of the slider,

$$v_p = \omega r \left(\sin \theta + \frac{\sin 2\theta}{2n} \right) = 47.13 \times 0.15 \left(\sin 60^\circ + \frac{\sin 120^\circ}{2 \times 4} \right) \text{ m/s}$$
$$= 6.9 \text{ m/s Ans.}$$

and acceleration of the slider,

$$a_p = \omega^2 r \left(\cos \theta + \frac{\cos 2\theta}{n} \right) = (47.13)^2 \times 0.15 \left(\cos 60^\circ + \frac{\cos 120^\circ}{4} \right) \text{ m/s}^2$$
$$= 124.94 \text{ m/s}^2 \text{ Ans.}$$

2. Angular velocity and angular acceleration of the connecting rod

We know that angular velocity of the connecting rod,

$$\omega_{PC} = \frac{\omega \cos \theta}{n} = \frac{47.13 \times \cos 60^\circ}{4} = 5.9 \text{ rad/s Ans.}$$

and angular acceleration of the connecting rod,

$$\alpha_{PC} = \frac{\omega^2 \sin \theta}{n} = \frac{(47.13)^2 \times \sin 60^\circ}{4} = 481 \text{ rad/s}^2 \text{ Ans.}$$

Module 3	Equivalent System
Lecture 6	Equivalent Dynamical System

Equivalent Dynamical System

In order to determine the motion of a rigid body, under the action of external forces, it is usually convenient to replace the rigid body by two masses placed at a fixed distance apart, in such a way that,

1. The sum of their masses is equal to the total mass of the body;
2. The centre of gravity of the two masses coincides with that of the body; and
3. The sum of mass moment of inertia of the masses about their centre of gravity is equal to the mass moment of inertia of the body.

When these three conditions are satisfied, then it is said to be an **equivalent dynamical system**.

Consider a rigid body, having its centre of gravity at G , as shown in Fig.

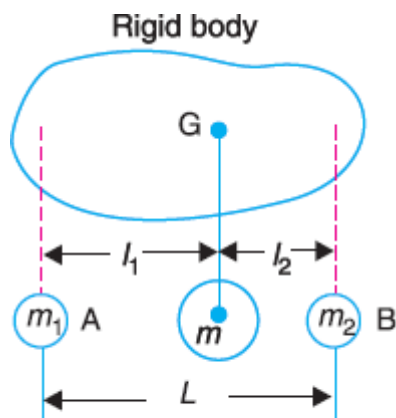


Fig. Equivalent dynamical system.

Let m = Mass of the body,

kG = Radius of gyration about its centre of gravity G ,

m_1 and m_2 = Two masses which form a dynamical equivalent system,

l_1 = Distance of mass m_1 from G ,

l_2 = Distance of mass m_2 from G , and

L = Total distance between the masses m_1 and m_2 .

Thus, for the two masses to be dynamically equivalent,

$$m_1 + m_2 = m \quad \dots(i)$$

$$m_1.l_1 = m_2.l_2 \quad \dots(ii)$$

$$\text{and} \quad m_1 (l_1)^2 + m_2 (l_2)^2 = m (k_G)^2 \quad \dots(iii)$$

From equations (i) and (ii),

$$m_1 = \frac{l_2.m}{l_1 + l_2} \quad \dots(iv)$$

$$\text{and} \quad m_2 = \frac{l_1.m}{l_1 + l_2} \quad \dots(v)$$

Substituting the value of m_1 and m_2 in equation (iii), we have

$$\frac{l_2.m}{l_1 + l_2} (l_1)^2 + \frac{l_1.m}{l_1 + l_2} (l_2)^2 = m (k_G)^2 \quad \text{or} \quad \frac{l_1 l_2 (l_1 + l_2)}{l_1 + l_2} = (k_G)^2$$

$$\therefore \quad l_1 l_2 = (k_G)^2 \quad \dots(vi)$$

This equation gives the essential condition of placing the two masses, so that the system becomes dynamical equivalent. The distance of one of the masses (*i.e.* either l_1 or l_2) is arbitrary chosen and the other distance is obtained from equation (vi).

Note : When the radius of gyration k_G is not known, then the position of the second mass may be obtained by considering the body as a compound pendulum. We have already discussed, that the length of the simple pendulum which gives the same frequency as the rigid body (*i.e.* compound pendulum) is

$$L = \frac{(k_G)^2 + h^2}{h} = \frac{(k_G)^2 + (l_1)^2}{l_1} \quad \dots(\text{Replacing } h \text{ by } l_1)$$

$$\text{We also know that} \quad l_1 l_2 = (k_G)^2$$

$$\therefore \quad L = \frac{l_1 l_2 + (l_1)^2}{l_1} = l_2 + l_1$$

This means that the second mass is situated at the *centre of oscillation or percussion* of the body, which is at a distance of $l_2 = (k_G)^2/l_1$.

Module 3	Equivalent System
Lecture 7	Problems on Equivalent System

Problems:

1. The connecting rod of a gasoline engines 300 mm long between its centres. It has a mass of 15 kg and mass moment of inertia of 7000 kg-mm². Its centre of gravity is at 200 mm from its small end centre. Determine the dynamical equivalent two-mass system of the connecting rod if one of the masses is located at the small end centre.

Given:

$$l = 300 \text{ mm}$$

$$m = 15 \text{ kg}$$

$$I = 7000 \text{ kg-mm}^2$$

$$l_1 = 200 \text{ mm}$$

To find:

$$m_1 = ?$$

$$m_2 = ?$$

Solution:

The connecting rod is shown in Fig. 15.16.

Let k_G = Radius of gyration of the connecting rod about an axis passing through its centre of gravity G .

We know that mass moment of inertia (I),

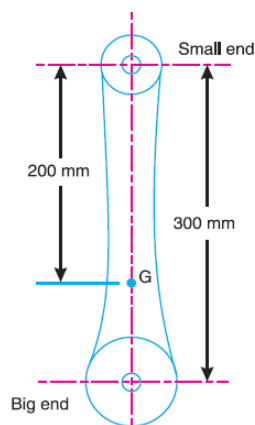


Fig. 15.16



Fig. 15.17

$$I = m K_G^2$$

$$K_G = 21.6 \text{ mm}$$

It is given that one of the masses is located at the small end centre. Let the other mass is placed at a distance l_2 from the centre of gravity G , as shown in Fig. 15.17.

We know that for a dynamical equivalent system,

$$l_1 l_2 = (K_G)^2$$

$$l_2 = 2.33 \text{ mm}$$

We know that,

$$m_1 = \frac{l_2 \cdot m}{l_1 + l_2} = 0.17 \text{ kg}$$

$$m_2 = \frac{l_1 \cdot m}{l_1 + l_2} = 14.87 \text{ kg}$$

2. A connecting rod is suspended from a point 25 mm above the centre of small end, and 650 mm above its centre of gravity, its mass being 37.5 kg. When permitted to oscillate, the time period is found to be 1.87 seconds. Find the dynamical equivalent system constituted of two masses, one of which is located at the small end centre.

Given:

$$h = 650 \text{ mm} = 0.65 \text{ m}$$

$$l_1 = 650 - 25 = 625 \text{ mm} = 0.625 \text{ m}$$

$$m = 37.5 \text{ kg}$$

$$tp = 1.87 \text{ s}$$

To find:

$$m_1 = ?$$

$$m_2 = ?$$

Solution:

First of all, let us find the radius of gyration (k_G) of the connecting rod (considering it is a compound pendulum), about an axis passing through its centre of gravity, G .

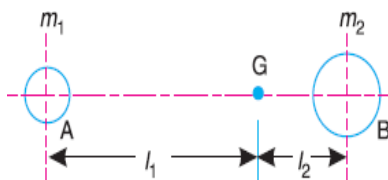


Fig. 15.19

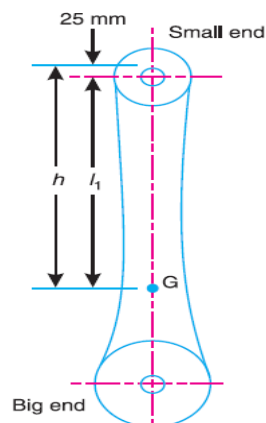


Fig. 15.18

We know that for a compound pendulum, time period of oscillation (tp),

$$t_p = 2\pi \sqrt{\frac{(K_G)^2 + h^2}{g \cdot h}}$$

$$K_G = 0.377 \text{ m}$$

We know that for a dynamical equivalent system,

$$l_1 l_2 = (K_G)^2$$

$$l_2 = 0.228 \text{ m}$$

We know that,

$$m_1 = \frac{l_2 \cdot m}{l_1 + l_2} = 10 \text{ kg}$$

$$m_2 = \frac{l_1 \cdot m}{l_1 + l_2} = 27.5 \text{ kg}$$

3. The following data relate to a connecting rod of a reciprocating engine:

Mass = 55 kg; Distance between bearing centres = 850 mm; Diameter of small end bearing = 75 mm; Diameter of big end bearing = 100 mm; Time of oscillation when the connecting rod is suspended from small end = 1.83 s; Time of oscillation when the connecting rod is suspended from big end = 1.68 s.

Determine: 1. the radius of gyration of the rod about an axis passing through the centre of gravity and perpendicular to the plane of oscillation; 2. the moment of inertia of the rod about the same axis; and 3. the dynamically equivalent system for the connecting rod, constituted of two masses, one of which is situated at the small end centre.

Given:

$$m = 55 \text{ kg}$$

$$l = 850 \text{ mm} = 0.85 \text{ m}$$

$$d_1 = 75 \text{ mm} = 0.075 \text{ m}$$

$$d_2 = 100 \text{ mm} = 0.1 \text{ m}$$

$$t_{p1} = 1.83 \text{ s}$$

$$t_{p2} = 1.68 \text{ s}$$

To find:

$$K_G = ?$$

$$I = ?$$

$$m_1 = ?$$

$$m_2 = ?$$

Solution:

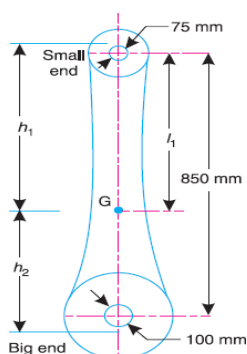


Fig. 15.20

1. Radius of gyration:

$$t_{p1} = 2\pi\sqrt{\frac{(K_G)^2 + h_1^2}{g \cdot h_1}}$$

$$K_G^2 = 0.832 h_1 - h_1^2 \text{-----} (1)$$

$$t_{p2} = 2\pi\sqrt{\frac{(K_G)^2 + h_2^2}{g \cdot h_2}}$$

$$K_G^2 = 0.70 h_2 - h_2^2 \text{-----} (2)$$

Equating (1) and (2)

$$0.832 h_1 - h_1^2 = 0.70 h_2 - h_2^2 \text{-----} (3)$$

From geometry

$$h_1 + h_2 = d_1/2 + l + d_2/2$$

$$h_1 + h_2 = 0.9375 \text{-----}(4)$$

Equating (3) and (4)

$$h_1 = 0.65 \text{ m}$$

$$K_G^2 = 0.1183$$

$$K_G = 0.343 \text{ m}$$

2. Moment of inertia:

$$I = m K_G^2 = 6.51 \text{ kg-m}^2$$

3. Dynamically equivalent system of the rod:

$$l_1 = h_1 - d_1/2 = 0.6125 \text{ m}$$

$$l_1 l_2 = (K_G)^2$$

$$l_2 = 0.193 \text{ m}$$

We know that,

$$m_1 = \frac{l_2}{l_1 + l_2} = 13.18 \text{ kg}$$

$$m_2 = \frac{l_1}{l_1 + l_2} = 41.82 \text{ kg}$$

Module 4	Turning Moment
Lecture 8	Turning Moment Diagram:

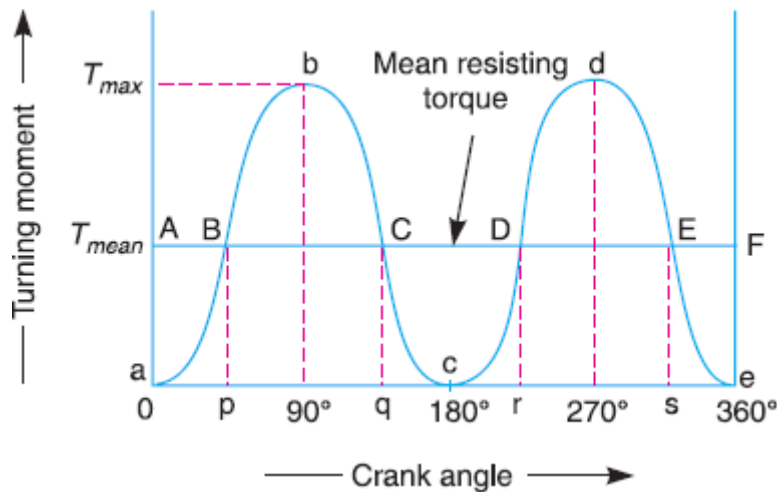
Turning Moment Diagram:

The turning moment diagram is graphical representation of the turning moment or crank effort for various positions of crank.

Single cylinder double acting engine:

A turning moment diagram for a single cylinder double acting steam engine is shown in Fig. The vertical ordinate represents the turning moment and the horizontal ordinate represents the crank angle. The turning moment on the crankshaft,

$$T = F_P \times r \left(\sin \theta + \frac{\sin 2\theta}{2\sqrt{n^2 - \sin^2 \theta}} \right)$$



Where

F_P = Piston effort,

r = Radius of crank,

n = Ratio of the connecting rod length and radius of crank, and

θ = Angle turned by the crank from inner dead centre.

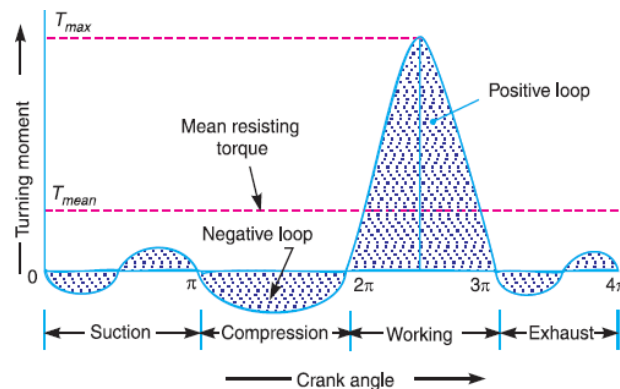
From the above expression, we see that the turning moment (T) is zero, when the crank angle (θ) is zero. It is maximum when the crank angle is 90° and it is again zero when crank angle is 180° . This is shown by the curve abc in Fig and it represents the turning moment diagram for outstroke.

The curve cde is the turning moment diagram for in stroke and is somewhat similar to the curve abc . Since the work done is the product of the turning moment and the angle turned, therefore the area of the turning moment diagram represents the work done per revolution.

In actual practice, the engine is assumed to work against the mean resisting torque, as shown by a horizontal line AF . The height of the ordinate aA represents the mean height of the turning moment diagram. Since it is assumed that the work done by the turning moment per revolution is equal to the work done against the mean resisting torque, therefore the area of the rectangle $aAF e$ is proportional to the work done against the mean resisting torque.

Turning moment diagram for 4-stroke I.C engine:

A turning moment diagram for a four stroke cycle internal combustion engine is shown in Fig. We know that in a four stroke cycle internal combustion engine, there is one working stroke after the crank has turned through two revolutions, *i.e.* 720° (or 4π radians).

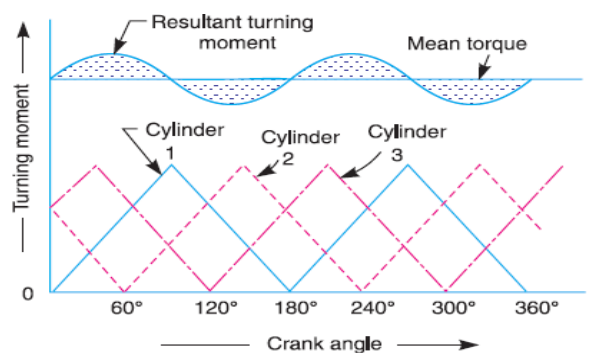


Turning moment diagram for a four stroke cycle internal combustion engine.

Since the pressure inside the engine cylinder is less than the atmospheric pressure during the suction stroke, therefore a negative loop is formed as shown in Fig. 16.2. During the compression stroke, the work is done on the gases, therefore a higher negative loop is obtained. During the expansion or working stroke, the fuel burns and the gases expand, therefore a large positive loop is obtained. In this stroke, the work is done by the gases. During exhaust stroke, the work is done on the gases, therefore a negative loop is formed. It may be noted that the effect of the inertia forces on the piston is taken into account in Fig.

Turning moment diagram for a multi cylinder engine

A separate turning moment diagram for a compound steam engine having three cylinders and the resultant turning moment diagram is shown in Fig. 16.3. The resultant turning moment diagram is the sum of the turning moment diagrams for the three cylinders. It may be noted that the first cylinder is the high pressure cylinder, second cylinder is the intermediate cylinder and the third cylinder is the low pressure cylinder. The cranks, in case of three cylinders, are usually placed at 120° to each other.

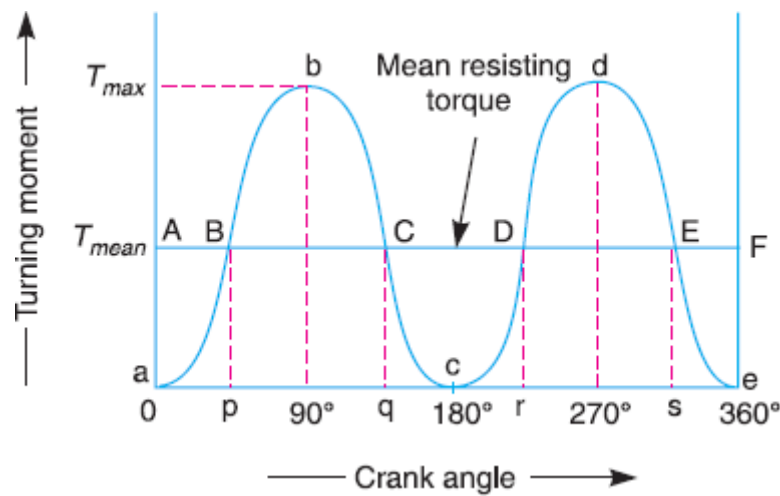


Turning moment diagram for a multi-cylinder engine.

Module 4	Turning Moment
Lecture 9	Fluctuation of Energy

Fluctuation of Energy

The difference in the kinetic energies at the point is called the maximum fluctuation of energy.



The fluctuation of energy may be determined by the turning moment diagram for one complete cycle of operation. Consider the turning moment diagram for a single cylinder double acting steam engine as shown in Fig. 16.1. We see that the mean resisting torque line AF cuts the turning moment diagram at points B, C, D and E . When the crank moves from a to p , the work done by the engine is equal to the area aBp , whereas the energy required is represented by the area $aABp$. In other words, the engine has done less work (equal to the area aAB) than the requirement. This amount of energy is taken from the flywheel and hence the speed of the flywheel decreases. Now the crank moves from p to q , the work done by the engine is equal to the area $pBbCq$, whereas the requirement of energy is represented by the area $pBCq$. Therefore, the engine has done more work than the requirement. This excess work (equal to the area BbC) is stored in the flywheel and hence the speed of the flywheel increases while the crank moves from p to q .

Similarly, when the crank moves from q to r , more work is taken from the engine than is developed. This loss of work is represented by the area CcD . To supply this loss, the flywheel gives up some of its energy and thus the speed decreases while the crank moves from q to r . As the crank moves from r to s , excess energy is again developed given by the area DdE and the speed again increases. As the piston moves from s to e , again there is a loss of work and the speed decreases. The variations of energy above and below the mean resisting torque line are called *fluctuations of energy*. The areas BbC, CcD, DdE , etc. represent fluctuations of energy. A little consideration will show that the engine has a maximum speed either at q or at s . This is due to the fact that the flywheel absorbs energy while the crank moves from p to q and from r to s . On the other hand, the engine has a minimum speed either at p or at r . The reason is that the flywheel gives out some of its energy when the crank moves from a to p and q to r . The difference between the maximum and the minimum energies is known as *maximum fluctuation of energy*.

Fluctuation of Speed

This is defined as the ratio of the difference between the maximum and minimum angular speeds during a cycle to the mean speed of rotation of the crank shaft.

Maximum fluctuation of energy:

A turning moment diagram for a multi-cylinder engine is shown by a wavy curve in Fig. The horizontal line AG represents the mean torque line. Let a_1, a_3, a_5 be the areas above the mean torque line and a_2, a_4 and a_6 be

the areas below the mean torque line. These areas represent some quantity of energy which is either added or subtracted from the energy of the moving parts of the engine.

Let the energy in the flywheel at $A = E$, then from Fig, we have

Energy at $B = E + a_1$

Energy at $C = E + a_1 - a_2$

Energy at $D = E + a_1 - a_2 + a_3$

Energy at $E = E + a_1 - a_2 + a_3 - a_4$

Energy at $F = E + a_1 - a_2 + a_3 - a_4 + a_5$

Energy at $G = E + a_1 - a_2 + a_3 - a_4 + a_5 - a_6$

= Energy at A (i.e. cycle repeats after G)

Let us now suppose that the greatest of these energies is at B and least at E . Therefore,

Maximum energy in flywheel

= $E + a_1$

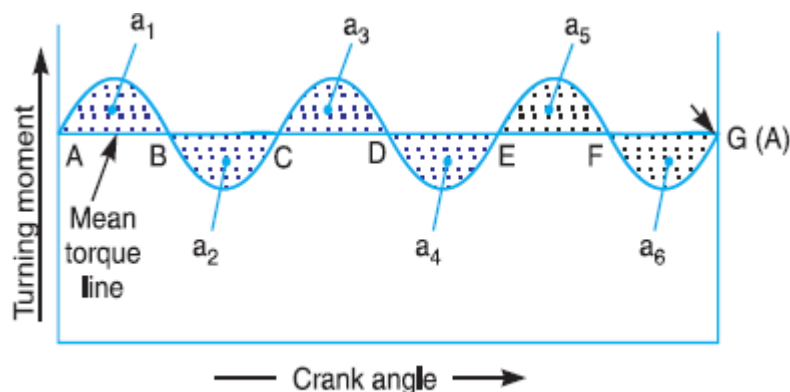
Minimum energy in the flywheel

= $E + a_1 - a_2 + a_3 - a_4$

∴ Maximum fluctuation of energy,

$\Delta E = \text{Maximum energy} - \text{Minimum energy}$

= $(E + a_1) - (E + a_1 - a_2 + a_3 - a_4) = a_2 - a_3 + a_4$



Determination of maximum fluctuation of energy.

(15) Coefficient of fluctuation of energy:

It may be defined as the ratio of the maximum fluctuation of energy to the work done per cycle.

Mathematically, coefficient of fluctuation of energy,

$$C_E = \frac{\text{Maximum fluctuation of energy}}{\text{Work done per cycle}}$$

The work done per cycle (in N-m or joules) may be obtained by using the following two relations : 1.

$$\text{Work done per cycle} = T_{mean} \times \theta$$

Where T_{mean} = Mean torque, and

θ = Angle turned (in radians), in one revolution.

= 2π , in case of steam engine and two stroke internal combustion engines

= 4π , in case of four stroke internal combustion engines.

The mean torque (T_{mean}) in N-m may be obtained by using the following relation:

$$T_{mean} = \frac{P \times 60}{2\pi N} = \frac{P}{\omega}$$

Where P = Power transmitted in watts,

N = Speed in r.p.m., and

ω = Angular speed in rad/s = $2\pi N/60$

2. The work done per cycle may also be obtained by using the following relation :

$$\text{Work done per cycle} = \frac{P \times 60}{n}$$

Where n = Number of working strokes per minute,

= N , in case of steam engines and two stroke internal combustion engines,

= $N/2$, in case of four stroke internal combustion engines.

The following table shows the values of coefficient of fluctuation of energy for steam engines and internal combustion engines.

Table . Coefficient of fluctuation of energy (C_E) for steam and internal combustion engines.

S.No.	Type of engine	Coefficient of fluctuation of energy (C_E)
1.	Single cylinder, double acting steam engine	0.21
2.	Cross-compound steam engine	0.096
3.	Single cylinder, single acting, four stroke gas engine	1.93
4.	Four cylinders, single acting, four stroke gas engine	0.066
5.	Six cylinders, single acting, four stroke gas engine	0.031

Coefficient of fluctuation of speed

The difference between the maximum and minimum speeds during a cycle is called the *maximum fluctuation of speed*. The ratio of the maximum fluctuation of speed to the mean speed is called the *coefficient of fluctuation of speed*.

Let N_1 and N_2 = Maximum and minimum speeds in r.p.m. during the cycle, and

$$N = \text{Mean speed in r.p.m.} = \frac{N_1 + N_2}{2} \quad C_s = \frac{N_1 - N_2}{N} = \frac{2(N_1 - N_2)}{N_1 + N_2}$$

$$= \frac{\omega_1 - \omega_2}{\omega} = \frac{2(\omega_1 - \omega_2)}{\omega_1 + \omega_2} \quad \dots(\text{In terms of angular speeds})$$

$$= \frac{v_1 - v_2}{v} = \frac{2(v_1 - v_2)}{v_1 + v_2} \quad \dots(\text{In terms of linear speeds})$$

Energy stored in flywheel:

A flywheel is a rotating mass that is used as an energy reservoir in a machine. It absorbs energy in the form of kinetic energy, during those periods of crank rotation when actual turning moment is greater than the resisting moment and release energy, by way of parting with some of its K.E, when the actual turning moment is less than the resisting moment.

Flywheel in punching press:

The flywheels used for prime movers constitute a class of problems in which the resisting torque is assumed to be constant and the driving torque varies. flywheels used in punching riveting and similar machines constitute another class of problems in which the actual(driving) turning moment provided by an electric motor is more or less constant but the resisting torque(load) varies.

Module 4	Turning Moment
Lecture 10	Problems on Turning moment

Problems:

1. The mass of flywheel of an engine is 6.5 tonnes and radius of gyration is 1.8 m. It is found from the turning moment diagram that the fluctuation of energy is 56 KN-m. If the mean speed of an engine is 120 r.p.m., find the maximum and minimum speeds.

Given:

$$m = 6.5 \text{ tonnes} = 6500 \text{ kg}$$

$$k = 1.8 \text{ m}$$

$$\Delta E = 56 \text{ KN-m} = 56 \times 10^3 \text{ N-m}$$

$$N = 120 \text{ r.p.m}$$

To Find:

$$N_1 = ?$$

$$N_2 = ?$$

Solution:

Fluctuation of energy

$$\Delta E = \frac{\pi^2}{900} \times m \cdot k^2 \times N (N_1 - N_2)$$

$$N_1 - N_2 = 2 \text{ ----- (1)}$$

Mean Speed

$$N = (N_1 + N_2)/2$$

$$N_1 + N_2 = 240 \text{ ----- (2)}$$

From equation (1) and (2)

$$N_1 = 121 \text{ r.p.m}$$

$$N_2 = 119 \text{ r.p.m}$$

2. The flywheel of a steam engine has a radius of gyration of 1 m and mass 2500 kg. The starting torque of the steam engine is 1500 N-m and may assumed constant. Determine the angular acceleration of the flywheel and the kinetic energy of the flywheel after 10 seconds from the start.

Given:

$$k = 1 \text{ m}$$

$$m = 2500 \text{ kg}$$

$$T = 1500 \text{ N-m}$$

To Find:

$$\alpha = ?$$

$$\text{K.E} = ?$$

Solution:

1. Angular acceleration of the flywheel:

Moment of inertia of the flywheel

$$I = m.k^2 = 2500 \text{ kg-m}^2$$

Starting torque of the engine (T)

$$T = I.\alpha$$

$$\alpha = 0.6 \text{ rad/ s}^2$$

2. Kinetic energy of the flywheel:

$$\omega_1 = \text{angular speed at rest} = 0$$

$$\omega_2 = \text{angular speed after 10 seconds}$$

$$t = \text{time in seconds}$$

$$\omega_2 = \omega_1 + \alpha t = 6 \text{ rad/s}$$

$$\text{K.E} = \frac{1}{2} \times I \times (\omega_2)^2$$

$$\text{K.E} = 45000 \text{ N-m} = 45 \text{ KN-m}$$

-
3. The horizontal cross compound steam engine develops 300 kW at 90 r.p.m. The coefficient of fluctuation of energy as found from the turning moment diagram is to be 0.1 and the fluctuation of speed is to be kept within $\pm 0.5\%$ of the mean speed. Find the weight of the flywheel required, if the radius of gyration is 2 meters.

Given:

$$P = 300 \text{ kW} = 300000 \text{ W}$$

$$N = 90 \text{ r.p.m}$$

$$C_E = 0.1$$

$$k = 2 \text{ m}$$

$$\text{Fluctuation of speed} = \pm 0.5\%$$

$$\text{Total fluctuation } (C_s) = 1\%$$

To Find:

$$m = ?$$

Solution:

$$\omega = 2\pi N / 60 = 9.426 \text{ rad/s}$$

Work done per cycle

$$= (P \times 60) / N$$

$$= 200000 \text{ N-m}$$

Maximum fluctuation of energy (ΔE) = W.D $\times C_E$

$$\Delta E = 20000 \text{ N-m}$$

We know that,

$$\Delta E = m.k^2. \omega^2.C_s$$

$$m = 5630 \text{ kg}$$

-
4. The turning moment diagram of a petrol engine is drawn to the following scales: turning moment 1mm = 5 N-m; crank angle, 1 mm = 1°. The turning moment diagram repeats itself at every half revolution of the engine and the area above and below the mean turning moment line taken in order are 295, 685, 40, 340, 960, 270 mm². The rotating parts are equivalent to a mass of 36 kg at a radius of gyration of 150 mm. Determine the coefficient of fluctuation of speed when the engine runs at 1800 r.p.m.

Given

$$m = 36 \text{ kg}$$

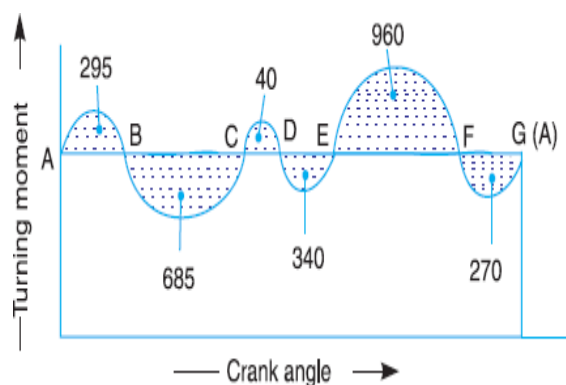
$$k = 150 \text{ mm} = 0.15 \text{ m}$$

$$N = 1800 \text{ r.p.m}$$

To Find:

$$C_s = ?$$

Solution:



$$\omega = 2\pi N / 60 = 188.52 \text{ rad/s}$$

Since the turning moment scale is $1\text{ mm} = 5 \text{ N-m}$

Crank angle scale is $1\text{ mm} = 1^\circ = \pi / 180 \text{ rad}$

$$1 \text{ mm}^2 \text{ on turning moment diagram} = 5 \times \frac{\pi}{180}$$

$$= \frac{\pi}{36} \text{ N-m}$$

Let the total energy at A = E

$$\text{Energy at B} = E + 295$$

$$\text{Energy at C} = E + 295 - 685 = E - 390$$

$$\text{Energy at D} = E - 390 + 40 = E - 350$$

$$\text{Energy at E} = E - 350 - 340 = E - 690$$

$$\text{Energy at F} = E - 690 + 960 = E + 270$$

$$\text{Energy at G} = E + 270 - 270 = E$$

$\Delta E = \text{maximum energy} - \text{minimum energy}$

$$= (E + 295) - (E - 690) = 985 \text{ mm}^2$$

$$= 985 \times \frac{\pi}{36} = 86 \text{ N-m}$$

We know that,

$$\Delta E = m.k^2 . \omega^2 . C_s$$

$$C_s = 0.003 = 0.3 \%$$

5. The turning moment diagram for a multi cylinder engine has been drawn to a scale $1 \text{ mm} = 600 \text{ N-m}$ vertically and $1\text{ mm} = 3^\circ$ horizontally. The intercepted areas between the output torque curve and the mean resistance line, taken in order from one end, are as follows: + 52, - 124, + 92, - 140, + 85, - 72 and + 107 mm^2 . When the engine is running at a speed of 600 r.p.m. if the total fluctuation of speed is not to exceed $\pm 1.5\%$ of the mean, find the necessary mass of the flywheel of radius 0.5 m.

Given:

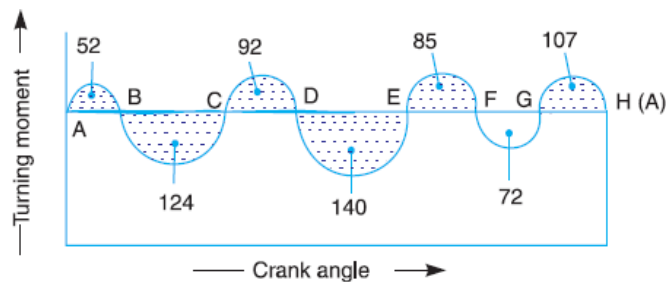
$$N = 600 \text{ r.p.m}$$

$$R = 0.5 \text{ m}$$

To Find:

$$m = ?$$

Solution:



$$\omega = 2\pi N / 60 = 62.84 \text{ rad/s}$$

Since the turning moment scale is $1\text{mm} = 600 \text{ N-m}$

Crank angle scale is $1\text{mm} = 3^\circ = 3\pi / 180 \text{ rad}$

$$1 \text{ mm}^2 \text{ on turning moment diagram} = 600 \times \frac{3\pi}{180}$$

$$= 31.42 \text{ N-m}$$

Fluctuation of speed = $\pm 1.5\%$

Total fluctuation (C_s) = 3%

Let the total energy at A = E

$$\text{Energy at B} = E + 52$$

$$\text{Energy at C} = E + 52 - 124 = E - 72$$

$$\text{Energy at D} = E - 72 + 92 = E + 20$$

$$\text{Energy at E} = E + 20 - 140 = E - 120$$

$$\text{Energy at F} = E - 120 + 85 = E - 35$$

$$\text{Energy at G} = E - 35 - 72 = E - 107$$

$$\text{Energy at H} = E - 107 + 107 = E$$

ΔE = maximum energy – minimum energy

$$= (E + 52) - (E - 120) = 172 \text{ mm}^2$$

$$= 172 \times 31.42 = 5404 \text{ N-m}$$

We know that,

$$\Delta E = m.k^2 \cdot \omega^2 \cdot C_s$$

$$m = 183 \text{ kg}$$

-
6. A shaft fitted with a flywheel rotates at 250 r.p.m and drives a machine. The torque of machine varies in a cyclic manner over a period of 3 revolutions. The torque rises from 750 N-m to 3000 N-m uniformly during $\frac{1}{2}$ revolution and remains constant for 1 revolution, the cycle being repeated thereafter. Determine the power required to drive the machine and percentage of fluctuation of speed, if the driving torque applied to the shaft is constant and the mass of the flywheel is 500 kg with radius of gyration of 600mm.

Given:

$$N = 250 \text{ r.p.m}$$

$$m = 500 \text{ kg}$$

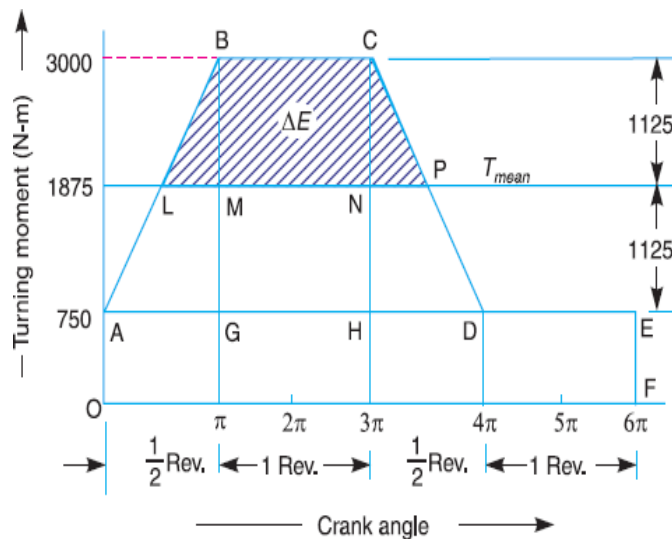
$$k = 0.6 \text{ m}$$

To Find:

$$P = ?$$

$$C_s = ?$$

Solution:



The turning moment diagram as shown in fig

$$\omega = 2\pi N / 60 = 26.2 \text{ rad/s}$$

Torque required for one complete cycle

$$= \text{area of figure OABCDEF}$$

$$= \text{Area OAEF} + \text{Area ABG} + \text{Area BCHG} + \text{Area CDH}$$

$$= OA \times OF + \frac{1}{2} \times AG \times BG + GH \times CH + \frac{1}{2} \times HD \times CH$$

$$= 6\pi \times 750 + \frac{1}{2} (3000-750) + 2\pi(3000 - 750) + \frac{1}{2} \pi(3000-750) T$$

$$= 11250 \pi \text{ N-m}$$

$$T_{\text{mean}} = T/6\pi = 1875 \text{ N-m}$$

1. Power required to drive the machine

$$P = T_{\text{mean}} \times \omega = 49.125 \text{ kW}$$

2. Coefficient of fluctuation of speed:

$$LM = 0.5\pi, MN = \pi, NP = 0.5\pi, BM = 1125 \text{ N-m}$$

$$\Delta E = \text{area of LBCP}$$

$$= \text{Area LBM} + \text{Area MBCN} + \text{Area PNC}$$

$$= \frac{1}{2} LM \times BM + MN \times BM + NP \times CN$$

$$= 8837 \text{ N-m}$$

$$\Delta E = m.k^2. \omega^2. C_s$$

$$C_s = 0.071$$

7. During forward stroke of the piston of the double acting steam engine, the turning moment has the maximum value of 2000 N-m when the crank makes an angle of 80° with the inner dead centre. During the backward stroke, the maximum turning moment is 1500 N-m when the crank makes an angle of 80° with the outer dead centre. The turning moment diagram for the engine may be assumed for simplicity to be represented by two triangles.

If the crank makes 100 r.p.m. and the radius of gyration of the flywheel is 1.75 m, find the coefficient of fluctuation of energy and the mass of the flywheel to keep the speed within $\pm 0.75\%$ of the mean speed. Also determine the crank angle at which the speed has its minimum and maximum values.

Given:

$$N = 100 \text{ r.p.m}$$

$$k = 1.75 \text{ m}$$

$$C_s = 1.5 \% = 0.015$$

To Find:

$$C_E = ?$$

$$m = ?$$

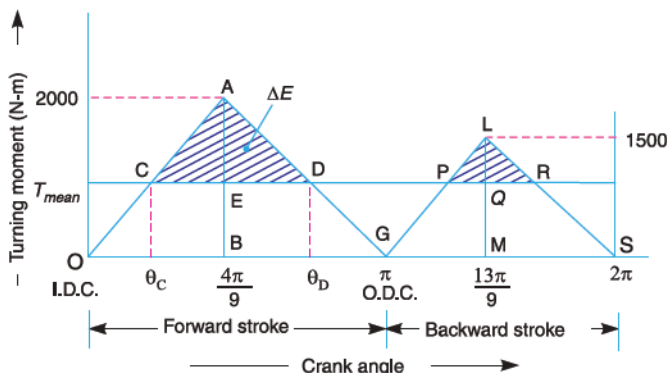
$$\text{Crank angle} = ?$$

Solution:

1. *Coefficient of fluctuation of energy*

The turning moment diagram for the engine during forward and backward strokes is shown in Fig. The point O represents the inner dead centre (I.D.C.) and point G represents the outer dead centre (O.D.C.). We know that maximum turning moment when crank makes an angle of 80° (or $80 \times \pi / 180 = 4\pi/9$ rad) with I.D.C.,

$$AB = 2000 \text{ N-m}$$



And maximum turning moment when crank makes an angle of 80° with outer dead centre (O.D.C.) or $180^\circ + 80^\circ = 260^\circ = 260 \times \pi / 180 = 13\pi/9$ rad with I.D.C.,

$$LM = 1500 \text{ N-m}$$

$$T_{mean} = EB = QM = \text{Mean resisting torque.}$$

We know that work done per cycle

$$= \text{Area of triangle } OAG + \text{Area of triangle } GLS$$

$$= \frac{1}{2} \times OG \times AB + \frac{1}{2} \times GS \times LM$$

$$= \frac{1}{2} \times \pi \times 2000 + \frac{1}{2} \times \pi \times 1500 = 1750 \pi \text{ N-m} \text{ ----- (i)}$$

We also know that work done per cycle

$$= T_{mean} \times 2 \pi \text{ N-m} \text{ ----- (ii)}$$

From equations (i) and (ii),

$$T_{mean} = 1750 \pi / 2 \pi = 875 \text{ N-m}$$

From similar triangles ACD and AOG ,

$$\frac{CD}{AE} = \frac{OG}{AB}$$

$$CD = 1.764 \text{ rad}$$

$$\Delta E = \text{Area of triangle } ACD = \frac{1}{2} \times CD \times AE$$

$$= \frac{1}{2} \times CD (AB - EB) = 992 \text{ N-m}$$

We know that coefficient of fluctuation of energy,

$$C_E = \frac{\text{Max. fluctuation of energy}}{\text{Work done per cycle}} = 0.18$$

2. Mass of flywheel

Let m = Mass of the flywheel.

We know that maximum fluctuation of energy (ΔE),

$$\Delta E = m.k^2. \omega^2. C_s$$

$$m = 197.2 \text{ kg}$$

3. Crank angle for minimum and maximum speed

We know that the speed of the flywheel is minimum at point C and maximum at point

Let θ_C and θ_D = Crank angles from I.D.C., for the minimum and maximum speeds.

From similar triangles ACE and AOB ,

$$\frac{CE}{OB} = \frac{AE}{AB}$$

$$CE = \pi/4 \text{ rad}$$

$$\theta_C = 35^\circ$$

Again from similar triangles AED and ABG ,

$$\frac{ED}{BG} = \frac{AE}{AB}$$

$$ED = 2.8\pi /9 \text{ rad}$$

$$\theta_C = 136^\circ$$

8. A three cylinder single acting engine has its cranks set equally at 120° and it runs at 600 r.p.m. The torque-crank angle diagram for each cycle is a triangle for the power stroke with a maximum torque of 90 N-m at 60° from dead centre of corresponding crank. The torque on the return stroke is sensibly zero. Determine: 1. power developed, 2. Coefficient of fluctuation of speed, if the mass of the flywheel is 12 kg and has a radius of gyration of 80 mm, 3. Coefficient of fluctuation of energy, and 4. Maximum angular acceleration of the flywheel.

Given:

$$N = 600 \text{ r.p.m. or } \omega = 2\pi \times 600/60 = 62.84 \text{ rad/s;}$$

$$T_{max} = 90 \text{ N-m;}$$

$$m = 12 \text{ kg;}$$

$$k = 80 \text{ mm} = 0.08 \text{ m}$$

To Find:

$$P = ?$$

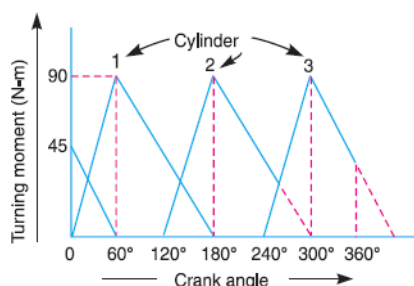
$$C_S = ?$$

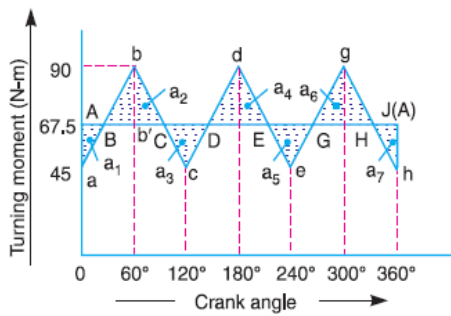
$$C_E = ?$$

$$\alpha = ?$$

Solution:

The torque-crank angle diagram for the individual cylinders is shown in Fig. (a), and the resultant torque-crank angle diagram for the three cylinders is shown in Fig. (b).





1. Power developed

We know that work done/cycle

$$\text{Area of three triangles} = 3 \times \frac{1}{2} \times \pi \times 90 = 424 \text{ N-m}$$

$$\text{And mean torque, } T_{\text{mean}} = \frac{\text{Work done / cycle}}{\text{Crank angle / cycle}}$$

$$= \frac{424}{2\pi} = 67.5 \text{ N - m}$$

$$\therefore \text{Power developed} = T_{\text{mean}} \times \omega = 67.5 \times 62.84$$

$$= 4240 \text{ W} = 4.24 \text{ kW Ans.}$$

2. Coefficient of fluctuation of speed

Let CS = Coefficient of fluctuation of speed.

First of all, let us find the maximum fluctuation of energy (ΔE).

From Fig. (b), we find that

$$\begin{aligned} a_1 &= \text{Area of triangle } AaB = \frac{1}{2} \times AB \times Aa \\ &= \frac{1}{2} \times \frac{\pi}{6} \times (67.5 - 45) = 5.89 \text{ N-m} = a_7 \quad \dots (\because AB = 30^\circ = \pi/6 \text{ rad}) \\ a_2 &= \text{Area of triangle } BbC = \frac{1}{2} \times BC \times bb' \\ &= \frac{1}{2} \times \frac{\pi}{3} (90 - 67.5) = 11.78 \text{ N-m} \quad \dots (\because BC = 60^\circ = \pi/3 \text{ rad}) \\ &= a_3 = a_4 = a_5 = a_6 \end{aligned}$$

Now, let the total energy at A = E, then referring to Fig. (b),

$$\text{Energy at B} = E - 5.89$$

$$\text{Energy at C} = E - 5.89 + 11.78 = E + 5.89$$

$$\text{Energy at D} = E + 5.89 - 11.78 = E - 5.89$$

$$\text{Energy at E} = E - 5.89 + 11.78 = E + 5.89$$

$$\text{Energy at G} = E + 5.89 - 11.78 = E - 5.89$$

$$\text{Energy at H} = E - 5.89 + 11.78 = E + 5.89$$

$$\text{Energy at J} = E + 5.89 - 5.89 = E = \text{Energy at A}$$

From above we see that maximum energy

$$= E + 5.89$$

And minimum energy = $E - 5.89$

∴ * Maximum fluctuation of energy,

$$\Delta E = (E + 5.89) - (E - 5.89) = 11.78 \text{ N-m}$$

We know that maximum fluctuation of energy (ΔE),

$$11.78 = m.k^2.\omega^2.CS = 12 \times (0.08)^2 \times (62.84)^2 \times CS = 303.3 \text{ CS}$$

$$\therefore CS = 11.78 / 303.3 = 0.04 \text{ or } 4\% \text{ Ans.}$$

3. Coefficient of fluctuation of energy

We know that coefficient of fluctuation of energy,

$$C_E = \frac{\text{Max fluctuation of energy}}{\text{Work done / cycle}} = \frac{11.78}{424} = 2.78 \%$$

4. Maximum angular acceleration of the flywheel

Let α = Maximum angular acceleration of the flywheel.

We know that,

$$T_{\max} - T_{\text{mean}} = I.\alpha = m.k^2.\alpha$$

$$\alpha = 292 \text{ rad/ S}^2$$

-
9. A single cylinder, single acting, four stroke gas engine develops 20 kW at 300 r.p.m. The work done by the gases during the expansion stroke is three times the work done on the gases during the compression stroke, the work done during the suction and exhaust strokes being negligible. If the total fluctuation of speed is not to exceed ± 2 per cent of the mean speed and the turning moment diagram during compression and expansion is assumed to be triangular in shape, find the moment of inertia of the flywheel.

Given:

$$P = 20 \text{ kW} = 20 \times 10^3 \text{ W};$$

$$N = 300 \text{ r.p.m. or } \omega = 2\pi \times 300/60 = 31.42 \text{ rad/s}$$

To Find

$$I = ?$$

Solution:

Since the total fluctuation of speed ($\omega_1 - \omega_2$) is not to exceed ± 2 per cent of the mean speed (ω), therefore

$$\omega_1 - \omega_2 = 4\% \omega$$

and coefficient of fluctuation of speed,

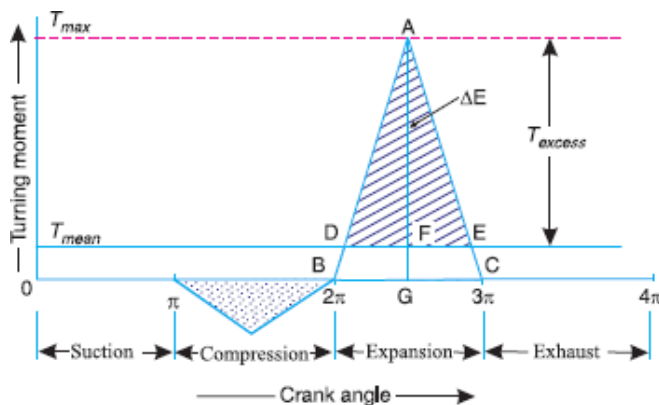
$$C_s = \frac{\omega_1 - \omega_2}{\omega} = 4\% = 0.04$$

The turning moment-crank angle diagram for a four stroke engine is shown in Fig... It is assumed to be triangular during compression and expansion strokes, neglecting the suction and exhaust strokes.

We know that for a four stroke engine, number of working strokes per cycle,

$$n = N/2 = 300 / 2 = 150$$

$$\therefore \text{Work done/cycle} = P \times 60/n = 20 \times 10^3 \times 60/150 = 8000 \text{ N-m}$$



Since the work done during suction and exhaust strokes is negligible, therefore net work done per cycle (during compression and expansion strokes)

$$= W_E - W_C = W_E - \frac{W_E}{3} = \frac{2}{3} W_E$$

Equating equations (i) and (ii), work done during expansion stroke,

$$W_E = 8000 \times 3/2 = 12\,000 \text{ N-m}$$

We know that work done during expansion stroke (W_E),

We know that work done during expansion stroke (W_E),

$$12\,000 = \text{Area of triangle } ABC = \frac{1}{2} \times BC \times AG = \frac{1}{2} \times \pi \times AG$$

$$AG = T_{max} = 12\,000 \times 2/\pi = 7638 \text{ N-m}$$

and mean turning moment,

$$*T_{mean} = FG = \frac{\text{Work done/cycle}}{\text{Crank angle/cycle}} = \frac{8000}{4\pi} = 637 \text{ N-m}$$

Excess turning moment,

$$T_{excess} = AF = AG - FG = 7638 - 637 = 7001 \text{ N-m}$$

Now, from similar triangles ADE and ABC ,

$$\frac{DE}{BC} = \frac{AF}{AG} \quad \text{or} \quad DE = \frac{AF}{AG} \times BC = \frac{7001}{7638} \times \pi = 2.88 \text{ rad}$$

Since the area above the mean turning moment line represents the maximum fluctuation of energy, therefore maximum fluctuation of energy,

$$\Delta E = \text{Area of } \triangle ADE = \frac{1}{2} \times DE \times AF = \frac{1}{2} \times 2.88 \times 7001 = 10081 \text{ N-m}$$

Let I = Moment of inertia of the flywheel in kg-m^2 .

We know that maximum fluctuation of energy (ΔE),

$$10\,081 = I \cdot \omega^2 \cdot CS = I \times (31.42)^2 \times 0.04 = 39.5 I$$

$$\therefore I = 10081 / 39.5 = 255.2 \text{ kg-m}^2 \text{ Ans.}$$

10. A multi-cylinder engine is to run at a speed of 600 r.p.m. On drawing the turning moment diagram to a scale of $1 \text{ mm} = 250 \text{ N-m}$ and $1 \text{ mm} = 3^\circ$, the areas above and below the mean torque line in mm^2 are : + 160, - 172, + 168, - 191, + 197, - 162. The speed is to be kept within $\pm 1\%$ of the mean speed of the engine. Calculate the necessary moment of inertia of the flywheel. Determine the suitable dimensions of a rectangular flywheel rim if the breadth is twice its thickness. The density of the cast iron is 7250 kg/m^3 and its hoop stress is 6 MPa . Assume that the rim contributes 92% of the flywheel effect.

Given:

$$N = 600 \text{ r.p.m. or } \omega = 2\pi \times 600/60 = 62.84 \text{ rad/s};$$

$$\rho = 7250 \text{ kg/m}^3;$$

$$\sigma = 6 \text{ MPa} = 6 \times 10^6 \text{ N/m}^2$$

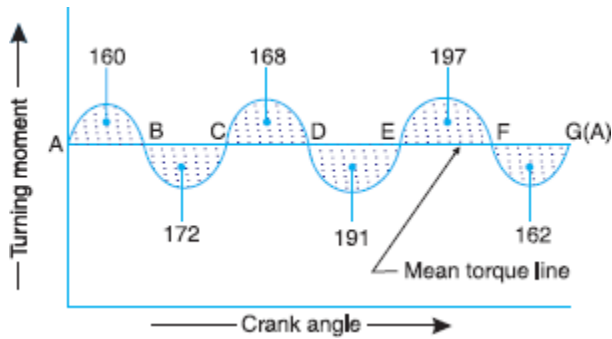
To Find:

$$I = ?$$

$$t = ?$$

b =?

Solution:



Since the fluctuation of speed is $\pm 1\%$ of mean speed, therefore, total fluctuation of speed, $\omega_1 - \omega_2 = 2\% \omega = 0.02 \omega$ and coefficient of fluctuation of speed,

$$C_s = \frac{\omega_1 - \omega_2}{\omega} = 0.02$$

Moment of inertia of the flywheel

Let I = Moment of inertia of the flywheel in kg-m².

The turning moment diagram is shown in Fig.. The turning moment scale is 1 mm = 250 N-m and crank angle scale is 1 mm = $3^\circ = \pi/60$ rad, therefore, 1 mm² of turning moment diagram = $250 \times \pi/60 = 13.1$ N-m

Let the total energy at $A = E$. Therefore from Fig. 16.19, we find that

$$\text{Energy at } B = E + 160$$

$$\text{Energy at } C = E + 160 - 172 = E - 12$$

$$\text{Energy at } D = E - 12 + 168 = E + 156$$

$$\text{Energy at } E = E + 156 - 191 = E - 35 \dots (\text{Minimum energy})$$

$$\text{Energy at } F = E - 35 + 197 = E + 162 \dots (\text{Maximum energy})$$

$$\text{Energy at } G = E + 162 - 162 = E = \text{Energy at } A$$

We know that maximum fluctuation of energy,

$$\Delta E = \text{Maximum energy} - \text{Minimum energy}$$

$$= (E + 162) - (E - 35) = 197 \text{ mm}^2$$

$$= 197 \times 13.1 = 2581 \text{ N-m}$$

We also know that maximum fluctuation of energy (ΔE),

$$2581 = I \omega^2 C_s = I \times (62.84)^2 \times 0.02 = 79 I$$

$$\therefore I = 2581/79 = 32.7 \text{ kg-m}^2 \text{ Ans.}$$

Dimensions of the flywheel rim

Let t = Thickness of the flywheel rim in metres,

b = Breadth of the flywheel rim in metres = $2 t$. . . (Given)

D = Mean diameter of the flywheel in metres, and

v = Peripheral velocity of the flywheel in m/s.

We know that hoop stress (σ),

$$6 \times 106 = \rho \cdot v^2 = 7250 v^2 \text{ or}$$

$$v^2 = 6 \times 106/7250 = 827.6$$

$$\therefore v = 28.8 \text{ m/s}$$

We know that $v = \pi D N/60$, or

$$D = v \times 60 / \pi N$$

$$= 28.8 \times 60 / \pi \times 600 = 0.92 \text{ m}$$

Now, let us find the mass (m) of the flywheel rim. Since the rim contributes 92% of the flywheel effect, therefore maximum fluctuation of energy of rim,

$$\Delta E_{rim} = 0.92 \times \Delta E$$

$$= 0.92 \times 2581 = 2375 \text{ N-m}$$

We know that maximum fluctuation of energy of rim (ΔE_{rim}),

$$2375 = m \cdot v^2 \cdot C_s = m \times (28.8)^2 \times 0.02 = 16.6 m$$

$$\therefore m = 2375/16.6 = 143 \text{ kg}$$

$$\text{Also } m = \text{Volume} \times \text{density} = \pi D \cdot A \cdot \rho = \pi D \cdot b \cdot t \cdot \rho$$

$$\therefore 143 = \pi \times 0.92 \times 2 t \times t \times 7250 = 41\,914 t^2$$

$$t^2 = 143 / 41\,914 = 0.0034 \text{ m}^2$$

$$\text{ort} = 0.0584 \text{ m} = 58.4 \text{ mm Ans.}$$

$$\text{and } b = 2 t = 116.8 \text{ mm Ans.}$$
